Department of Multidisciplinary Studies MDM-9

| MDM: Generative AI | B. Tech.: Civil/Comp/CompR/E&Tc/Mech/IT | | | | | Semester: VI | |
|---|---|---|---|---|---|---|---|
| **Course:** | **Generative AI Applications** | | | | | **Code:** | **BCS26MD05** |
| **Credits** | **Teaching Scheme (Hrs./Week)** | | | **Evaluation Scheme and Marks** | | | |
| | **Lecture** | **Practical** | **Tutorial** | **FA** | | **SA** | **Total** |
| | | | | **FA1** | **FA2** | | |
| **02** | **02** | **-** | **-** | **10** | **10** | **30** | **50** |

**Prior knowledge of:** Calculus, Linear Algebra, Probability Theory, and Python programming, are essential.

**Course Objectives:**

This course aims at enabling students,

1. To analyze the impact of GenAI in business
2. Understand ethical implications of Generative AI in science.
3. Analyze practical case studies for real-world insights.

**Course Outcomes:**

After learning the course, the students should be able to:

1. To understand the impact of generative AI in various fields.
2. Apply the concept of Gen AI in finance.
3. Apply Generative AI for simulation and modeling in various scientific fields.
4. Recognize the importance of ethical algorithms for fairness in AI systems.

**Detailed Syllabus**

| Unit | Description | Duration [Hrs] |
|---|---|---|

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|---|---|---|
| I | **Introduction**<br><br>The Impact of Generative AI on Business and Society, Emerging Trends and Technologies in Generative AI, Ethical Challenges and Regulatory Frameworks, Real-World Applications of Large Language Models, and Challenges and Limitations of Current Approaches. | 08 |
| II | **Generative AI in Finance and Business**<br><br>Algorithmic Trading and Financial Forecasting, Fraud Detection and Risk Management Customer Service and Chatbots, Marketing and Advertising Creativity, Business Process Automation and Optimization | 08 |
| III | **Generative AI in Science and Research**<br><br>Scientific Discovery and Hypothesis Generation<br><br>Simulation and Modeling in Various Fields, Data Augmentation and Synthesis, Collaboration and Knowledge Sharing, and Ethical and Social Implications of Generative AI. | 07 |
| IV | **Responsible AI**<br><br>Importance of ethical and unbiased algorithms in AI systems, Ethical frameworks and principles for AI development, Techniques for Implementing Unbiased Algorithms, Roles and responsibilities of stakeholders in the AI ecosystem. Case Study : IBM's AI Fairness 360 Toolkit | 07 |
| **Total** | | **30** |

**Text Books:**

1. Generative AI in Practice: 100+ Amazing Ways Generative Artificial Intelligence is Changing Business and Society, Bernard Marr

2. Responsible AI: Implementing Ethical and Unbiased Algorithms, by Shashin Mishra and Sray Agarwal

**-sources:**

1. https://elearn.nptel.ac.in/shop/iit-workshops/completed/leveraging-generative-ai-for-teaching-programming-cours es/?v=c86ee0d9d7ed

2. https://elearn.nptel.ac.in/shop/iit-workshops/completed/introduction-to-language-models/?v=c86ee0d9d7ed